

B. Tech Degree VIII (Supplementary) Semester Examination September 2006

ME 802 MACHINE DESIGN II

(2002 Admissions)

Time : 3 Hours

Maximum Marks : 100

(Use of Design Data Book is permitted)
(Assume reasonable data wherever necessary)

- I. (a) What are the advantages of band brake over block brake? (4)
 (b) What are the advantages of conical clutch over a plate clutch. Give its limitations. (4)
 (c) Design a suitable clutch for a 150cc scooter. Axial thrust is to be provided by suitable helical compression springs. Scooter develops a maximum power of 5.8HP at 5500rpm. At maximum torque point it develops 4HP at 3200rpm. Space restricts the outer diameter of the clutch of 150mm. (17)
- OR**
- II. (a) What are the different modes of chain drive failure? (4)
 (b) Define the term lay with reference to wire ropes. What are the different lays used in construction of wire ropes? (4)
 (c) Design a V belt drive to transmit 50KW at 1440rpm from an electric motor to a textile machine running 24hrs a day. The speed of machine shaft is 480rpm. (17)
- III. (a) With a neat sketch, explain the terms used in a spur gear. (5)
 (b) What are the advantages of helical gear over spur gear? (3)
 (c) Design a spur gear drive to transmit 30KW at 1440 rpm. Speed reduction is 2.5. The gear and pinion is made up of C 45 steel and the pressure angle is 20° . (17)
- OR**
- IV. (a) Differentiate between external bevel, internal bevel and crown gears. (4)
 (b) Explain the term self locking in a worm gear drive. (4)
 (c) Design a bevel gear drive between two shafts whose axes are at right angles. Speed of pinion shaft is 240rpm and that of the gear shaft is 120 rpm. Pinion is to have 21 teeth of involute profile with module of 20mm and a pressure angle of 20° and is to be of suitable material. Gear is of cast iron. Power at gear shaft is 75KW. (17)
- V. (a) State and explain fluid film lubrication. (5)
 (b) State the modes of failure of rolling contact bearing. (3)
 (c) Design a journal bearing for a centrifugal pump to the following specifications
 Diameter of journal = 75mm
 Speed of the journal = 1140rpm
 Load on each journal. = 11500 N (17)
- OR**
- VI. (a) Sketch the following bearings and give nomenclature
 (i) Self aligning ball bearing (5)
 (ii) Thrust bearing (5)
 (b) What are the relative merits and demerits of sliding contact bearing over rolling contact bearing? (5)
 (c) Select a suitable bearing which is to operate at 1500rpm and is acted upon by a 8000N radial load and 5000N thrust load. The inner ring rotates, the load is steady and the service is continuous. The shaft diameter is 45mm and life expectancy is 500 hrs. If the bearing outer ring is rotating and is subjected to light shock, how the life of bearing changes. (15)
- VII. (a) Enumerate general design recommendations for rolled sections. (10)
 (b) How working drawings are prepared for manufacturing. (7)
 (c) What are the design consideration for turned parts? (8)
- OR**
- VIII. (a) Explain with examples, how the surface finish is represented in working drawings. (7)
 (b) Explain the term tolerance with examples. (8)
 (c) Give the design recommendations for forged sections. (10)

